

Anower Hossen Zihad

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Research Assistant studying efficient distributed deep learning and beyond-backpropagation training. Combines method implementation, experimental design, and scientific writing with production AWS and Kubernetes experience.

Education

Montclair State University

Master of Computer Science; GPA: 4.00/4.00

Montclair, NJ

Expected Dec. 2026

Chittagong Independent University

Bachelor of Computer Science and Engineering; GPA: 3.90/4.00

Chittagong, Bangladesh

Sep. 2018 – Aug. 2022

Research Experience

Montclair State University

Research Assistant, School of Computing

Montclair, NJ

Aug. 2025 – Present

Research supervisor: Chao Huang

- Design experiments in beyond-backpropagation and distributed deep learning, investigating split learning, auxiliary losses, and graph neural networks (GNNs).
- Implement learning methods and training pipelines in **PyTorch**; run GPU experiments on university high-performance computing (HPC) systems and Google Colab across image-classification and graph-learning benchmarks.
- Design and analyze auxiliary-loss split-learning experiments on CIFAR-10/100, reporting **50% lower activation/gradient communication** and **up to 58% lower peak memory usage** versus standard split learning, with comparable predictive performance.
- Investigate knowledge distillation to mitigate forgetting in split learning; evaluate accuracy, convergence, runtime, communication, and memory.
- Formulate research questions, review literature, analyze results, and write and revise manuscripts with research collaborators.
- Presented research paper at the ICCN Workshop, IEEE INFOCOM 2026, Tokyo, Japan.

Publications and Preprints

1. Decoupled Split Learning via Auxiliary Loss

Anower Zihad, Felix Owino, Ming Tang, Chao Huang. arXiv:2601.19261, 2026.

2. Bangla Fake News Detection using Machine Learning, Deep Learning and Transformer Models

Risul Islam Rasel, Anower Hossen Zihad, et al. IEEE ICCIT, 2022, pp. 959–964.

Manuscripts Under Review and in Preparation

1. Tackling Forgetting via Knowledge Distillation: A Case Study in Split Learning

Under review, IEEE ICNC 2027.

2. Split Learning of GNN in a Beyond-Backpropagation Framework

In preparation for IEEE ICC 2027; extended study of the arXiv preprint above.

Technical Skills

ML and scientific computing: PyTorch, TensorFlow/Keras, NumPy, pandas; GPU experimentation, HPC

Programming: Python, SQL, Bash, Java, JavaScript

Cloud and infrastructure: AWS (EC2, VPC, S3, RDS, IAM, ECR, EKS), Docker, Kubernetes, Terraform, Ansible

Delivery and monitoring: Git, GitLab CI/CD, Jenkins, GitOps, Prometheus, Grafana

Industry Experience

TS4U LLC

DevOps Engineer

Warren, MI (Remote)

Feb. 2023 – Apr. 2024

- Deployed and monitored production microservices on AWS with Docker and Kubernetes; managed cloud networking, compute, storage, databases, and access controls.
- Automated infrastructure provisioning with Terraform and Ansible, reducing **provisioning time by 30%**.
- Implemented CI/CD workflows with GitLab CI/CD, Jenkins, and GitOps; integrated Prometheus, Grafana, and EFK monitoring and logging to support troubleshooting.